Supplementary Material

Introduction

The algorithms and data sets of QLARM are validated for the study area by matching the reported intensities and numbers of fatalities with theoretical calculations for the 13 earthquake in Table 1. We present this validation work in a supplemental section to shorten the body of the article. The method used in each case is the same, thus separation from the main text is in order. This *Supplementary Material* includes the validations of the 2003 Zhaosu, the 2009 Haixi, the 2012 Xinyuan-Hejing, the 2013 Minxian-Zhangxian, the 2014 Ludian, the 2015 Pishan and the 2016 Zaduo earthquakes, the 2010 Yushu and the 2016 Arketao earthquakes modeled as the line sources. The special cases of the 2003 Dayao, the 2008 Wenchuan and the 2014 Jinggu earthquakes are discussed in the main text.

The Zhaosu earthquake of 1 December 2003

For the 1 December 2003, Zhaosu earthquake, located near the border between Xinjiang, China and Kazakhstan, the magnitude was M_S 6.1/M_W 6.0, released by CEA and USGS, respectively. The CEA has published the intensity isoseismal map for the Chinese territory (Fig. 9.1; Shen et al., 2005). The I_{max} on Chinese territory is VIII and the maximum isoseismals form a narrow ellipse. We have no information of damage and casualties in Kazakhstan. Therefore, the reported numbers of fatalities and injured listed in Table 3 are minima, without those that may have occurred in Kazakhstan.

We used a line source (end-points are 42.94°N, 80.54°E and 42.99°N, 80.40°E) in this case, although it is a small earthquake because the isoseismals are elongated in the ESE direction.

For hypocentral depth, we first used that given by the CENC and also the moment tensor solution of the USGS, however, we had to refine the depth according to the observed intensities. The final I_{max} equals the reported one, and distribution of isoseismals of QLARM match the reported ones within a deviation of 20 km when the depth of the hypocenter is assumed to have been 21 km with a line source 14 km long and C_3 = 4.4 (Fig. 9.1). The length of the line source cannot be made much longer than that, because it corresponds to *M* 6 for average stress drop (Wyss, 1979; Wells & Coppersmith, 1994). We moved the epicenter from that of the CENC to the center of the isoseismals, and the distance between them is about 2 km.

Because there is no settlement near the epicenter, I_{max} is poorly defined and a difference in depth changes the I_{max} only little. The result shows that the I_{max} (calculated) is VIII, matching the observed value, and the ranges of the estimated number of fatalities and injured are 10-110 and 40-360, respectively (Table 3).

The Haixi earthquake of 28 August 2009

For the 28 August 2009, Haixi M_S 6.4/M_W 6.3 earthquake, located in the center of Qinghai Province, the CEA has published the intensity map (Fig. 9.1) and released the numbers of losses (see *Data and Resources*). The I_{max} is VII and because few people live in the area of the epicenter and nobody died. We have no information about injured people. Therefore, the reported numbers of fatalities and injured listed in Table 3 are zero and unknown, respectively. The unknown number of injured people suggests that nobody got hurt seriously in the 2009 Haixi earthquake.

The isoseismal map of the Haixi earthquake is elongated in an ESE direction, therefore the match using a line source is slightly better than that using a point source. A topographic line going through the epicenter with a strike equal to the elongation of the macro-seismic map, namely the direction of ESE, suggests that this may have been the direction of the fault generating this earthquake. Assuming for the Haixi earthquake an approximate length of 18 km for the line source (end-points are 37.67°N, 95.70°E and 37.62°N, 95.87°E, Table 1) is reasonable. The epicenter had to be moved significantly, with the distance between the epicenter from the CENC and the macro-epicenter being about 21 km.

We first used the hypocentral depth given by the CENC and the moment tensor solution of the USGS, then refined that according to the observed intensities. The final I_{max} equals the reported one and distribution of intensities of QLARM fit the reported ones within a deviation of 10 km, when the depth is 20 km with a rupture length of 18 km (Fig. 9.1). We need to get the strong intensities right because that is where fatalities occur. To match the lower intensities is less important.

The range of the estimated number of fatalities and injured is 0-30 and 10-120, respectively (Table 3). The number of injured cannot be compared with the reported ones because the latter is unknown.

The Yushu earthquake of 13 April 2010

The isoseismal map (Fig. 9.1) and losses (Table 3) for the 13 April 2010, M_S 7.3/M_W 6.9 Yushu earthquake in Qinghai province have been released by the CEA (see *Data and Resources*). The earthquake happened at about 8 a.m. local time, and caused ruinous damage in its meizoseismal region (I_{max} is IX, Fig. 9.1). 2,968 people were reported dead and at least 10,701 were injured (Table 3).

According to the slip model from Zhang et al (2010), we simplified the source as a rectangle considering the asperities that defined the rupture as having a length of 30 km, a width of 20 km and a dip of 80° (Table 2). For balancing the match of the intensities, we had to move the line source to the NW (end-points are given in Table 2). We place the epicenter in the middle of the meizoseismal area. The distance between the epicenter given by the CENC and the one we used is about 21 km.

First, we considered the hypocentral depth given by the CENC and also the moment tensor solution of the USGS. Then, we had to refine the depth according to the observed intensities. When we used a depth of 15 km along with an average attenuation $C_3 = 3.5$, the numbers of fatalities and I_{max} are about correct. We accepted the match presented in Figure 9.1 because we cannot improve it much. The results of Table 3 shows that the estimated fatalities fit well the reported ones and the slight underestimation of the number of injured is acceptable.

The Xinyuan-Hejing earthquake of 29 June 2012

For the Xinyuan-Hejing M_S 6.6/M_W 6.3 earthquake of 29 June 2012 in Xinjiang Uygur Autonomous Region, the isoseismal map (Fig. 9.1) and losses (Table 3) have been published by the CEA (see *Data and Resources*). Since few people live in that region, the macro-seismic information is marginal and this earthquake did not cause any serious casualties although it struck at about 5 a.m. local time. Nevertheless, this earthquake caused strong damage in its meizoseismal area (I_{max} from Fig. 9.1 is VIII).

We finally arrived at the optimal parameters for length of the line source, depth and attenuation, which meet the criteria we set before. Assuming approximately a length of 18 km for the line source (end-points are 43.34ºN, 84.60ºE and 43.30ºN, 84.47ºE) is reasonable. The final I_{max} equals the reported one and distribution of intensities computed by QLARM fit the reported ones within 5 km when the depth is 25 km with a lower attenuation (better transmission, here C_3 is 4.5). The necessary shift of the epicenter from the CENC to the macroepicenter is approximate 12 km.

The estimated numbers of calculated fatalities and injured, which are 30-350 and 130-1,000, respectively (Table 3), exaggerated the death toll slightly. The Xinyuan-Hejing earthquake happened early Saturday morning local time. The reported number of casualties luckily was low because local government began to implement a rural earthquake-resistant projects since 2004.

The Minxian-Zhangxian earthquake of 21 July 2013

For the Minxian-Zhangxian M_S 6.7/M_W 5.9 earthquake of 21 July 2013 in Gansu province, the isoseismal map (see *Data and Resources*) is shown in Figure 9.1 along with the intensities calculated by QLARM. The I_{max} of the reported isoseismals is VIII. This earthquake happened in the morning. Both, the high intensity in the meizoseismal area and the unfavorable origin time are mainly responsible for the numerous casualties.

The isoseismals are elongated, thus we constructed a line source (end-points are 34.57ºN, 104.09ºE and 34.52ºN, 104.18ºE) of 10 km length, corresponding to M_W 5.9. We moved the epicenter from the instrumental point to the center of the isoseismals by about 7 km. Assuming a lower than average attenuation (C_3 = 4.5) is sound for this case because this way the far-field is matched better (Fig. 9.1). However, we had to assume a depth of 28 km for balancing the intensity values at the center. In our final model the calculated intensities may still be too low, but the numbers of fatalities and injured (130-2,000 and 640-6,000, respectively) fit well.

The Ludian earthquake of 3 August 2014

For the Ludian M_S 6.6/M_W 6.2 earthquake of 3 August 2014 in Yunnan Province, the I_{max}(CEA) was IX (Fig. 9.1), which indicates 40%-70% of the buildings were badly damaged or destroyed (GB/T 17742-2008). 617 people were reported dead and at least 3,143 were injured (Table 3).

A line source of 15 km length was used for the calculation of intensities, and the distance that we moved the epicenter from the CENC to the center of the isoseismals is about 6 km (Fig. 9.1). We assume that the end-point coordinates of this line source are 27.14°N, 103.35°E and 27.01°N, 103.41°E. We used a lower than average attenuation, namely C_3 = 4.1, to increase the distant intensities. However in that case I_{max} also gets higher, so we need to reduce the depth to 12 km for balance at the center near I_{max}. The calculated ranges of the number of fatalities and injured are 190-1,000 and 650-4,000, respectively (Table 3) which fits the reported values.

The Pishan earthquake of 3 July 2015

For the Pishan M_S 6.5/M_W 6.4 earthquake of 3 July 2015 in Xinjiang Uygur Autonomous Region, Figure 9.1 shows the isoseismal from CEA along with the calculated intensities. The reported I_{max} was VIII. Three people were reported dead and at least 260 were injured (Table 3) based on the disaster assessment notice of the website of CEA (see *Data and Resources*).

We used the epicenter from CENC to calculate losses because that epicenter is located near the macroepicenter. The isoseismals are elongated in ESE direction. Therefore the final calculation was done with a line source (end-points are 37.60°N, 78.05°E and 37.52°N, 78.25°E) of 22 km length, corresponding to M_W 6.4. The fit looks good when we increase the focal depth from the ones of CENC and USGS to 20 km and use C_3 = 4.0 (Fig. 9.1). Table 3 shows that the estimated ranges of casualties are 50-480 and 190-2,000, respectively, which overestimated the reported numbers to an acceptable amount.

The Zaduo earthquake of 17 October 2016

For the 17 October 2016, Zaduo M_S 6.3/M_W 5.9 earthquake, located in Qinghai Province, the isoseismal map (Fig. 9.1) and the losses (see *Data and Resources*) have been published by the CEA. I_{max} was VII and because few people live in that area only one person died. The number of injured people has not been reported. The unknown number of injured people means that nobody got hurt seriously in this earthquake.

A line source of 10 km length, the end-point coordinates are 32.86ºN, 94.92ºE and 32.8ºN, 95.01ºE, was used for the calculation of intensities, and the distance that we moved the epicenter of CENC to the macroepicenter is about 9 km (Fig. 9.1). We used an average attenuation, namely C_3 = 3.5, for the calculation. The I_{max} equals the reported one and distribution of isoseismals of QLARM match the reported ones within 10 km when the depth of the hypocenter is assumed to have been 15 km. The calculation result shows that the ranges of the number of fatalities and injured are 0-10 and 0-40, respectively (Table 3), which agrees with the observations.

The Arketao earthquake of 25 November 2016

For the 25 November 2016, Arketao M_S 6.8/M_W 6.6 earthquake, located near the borders among Xinjiang of China, Kyrghyzstan and Tajikistan, we collected the information about the isoseismal map (Fig. 9.1) and investigation for losses (Table 3) in the disaster assessment notice of the website of CEA (see *Data and Resources*). The isoseismal map with I_{max} equals VIII is based on a field investigation organized by CEA, so the isoseismal map is accurate in the Chinese territory. The part of the isoseismals of Kyrghyzstan and Tajikistan is a rough interpolation. Information of damage and casualties in Kyrghyzstan and Tajikistan is not available. Hence, the reported numbers of casualties listed in Table 3 are minima, without those that may have occurred outside China.

We used a line source model which has been calculated by Zhang et al., then released by the Institute of Geophysics (IGP), CEA (end-points are 39.27ºN, 73.99ºE and 39.18ºN, 74.38ºE). The simplified source rectangle used is based on asperities with length of 30 km, width of 10 km and dip of 80º.

The final I_{max} equals the reported one and distribution of isoseismals of QLARM match the reported ones within 30 km when the depth is assumed to have been 19 km, with an average attenuation C_3 = 3.5. We believe this depth is appropriate because very shallow parts of the crust radiate little energy, even if there are some displacements, due to the shallow stress levels being low. We adjusted the center of the line source into the center of the felt area, and the distance from the instrumental epicenter estimate is about 7 km.

The calculated close-in intensities are well matched, far-out ones not so well. However, we should not increase the assumed transmission because we would estimate too many fatalities. Because few people live in the meizoseismal area only one person died. The ranges of the calculated number of fatalities and injured, 10-170 and 60-600, respectively (Table 3), overestimated the reported numbers to a reasonable amount.

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Appendix: The QLARM code

The procedures of the QLARM software to estimate fatalities and injured due to an earthquake are the following:

(1) The input to QLARM consists of epicenter, depth, magnitude, origin time, and a choice of several attenuation relationships, based on which intensities at all settlements in the QLARM data set are calculated (Trendafiliski et al., 2011). Instead of an epicenter point, a line source can be defined by entering the coordinates of the two endpoints. In that case, the shortest distance to this line is used for calculations of the strong ground motion. Because we must calculate intensities for comparison with observations, we cannot use attenuation relationships that yield peak ground accelerations or peak ground velocities. For calculating intensities we use equation (1) (Shebalin, 1985), in which case the three constants, C_i, defining the regional transmission properties of seismic waves, must be supplied.

$$I = C_1 M - C_2 \log\sqrt{r^2 + Z^2} + C_3 \tag{1}$$

in which, I is intensity in MMI scale, M is magnitude, r is distance from the epicenter, or line source, to the settlement and Z is depth, the latter two parameters in km. For the constants C_1, C_2 and C_3, we use the most common ones of 1.5, 4.5 and 3.5, respectively, unless different values are specified.

In QLARM, a settlement is a location of human habitation with a name, coordinates and its population. The population in most cases stems from the last census of the country in question (Trendafiliski et al., 2011). When there is no new census for several years, the population number in each settlement is increased by the average national growth of the population. In some countries, the populations of large settlements are available only. For the small settlements only name and coordinates are available from GeoNames (http://www.geonames.org/, last accessed 4 April 2018), thus we assume that each of these settlements contains that number of people which by its sum, plus the population in the known large settlements, yields the known total population of the country in question.

Each settlement is defined by a point. QLARM contains information on 239 countries, which include 1.93 million settlements (Trendafiliski et al., 2011). For about 70 large cities, the information on population and other properties are broken down into sections, which usually are defined as administrative districts. In models of large cities, soil conditions are considered by an amplification factor for each district.

(2) QLARM uses the damage grade scale (D_0-no damage; D_1-slight damage; D_2-moderate damage; D_3-heavy damage; D_4-very heavy damage; D_5-destruction) to classify the expected damage degree for each vulnerability class of the building stock. Considering the percentage of each building vulnerability class, the mean damage (*MD*; Trendafiliski et al., 2011) is calculated for each settlement, using

$$MD = 2.5[1 + \tanh(\frac{I + 6.25V_I - 13.1}{2.3})] \quad (2)$$

where V_I defines the membership of a particular building category in a specific vulnerability class. The membership is not deterministic, it defines the most likely class and its plausible and ultimate bounds. The occurrence probabilities of damage grade D_i in seismic intensity I_J are then beta-distributed according to the ranges of the mean damage grade (Trendafiliski et al., 2011). Therefore, QLARM creates a damage probability matrix for a particular vulnerability class.

(3) The final step is to calculate the possible human loss based on the damage degree of the building stock. The injury severity scale used is that given by HAZUS (c_1-non-injured; c_2-slightly injured; c_3-moderately injured; c_4-seriously injured; c_5-dying or dead; Federal Emergency Management Agency, 2003). This distribution is modified in QLARM and the original text can be found in Trendafiliski et al. (2011). As default values we use the HAZUS 2002 indoor casualty rates for building types corresponding to EMS-98 vulnerability classes (Table 4), which have been modified based on observed fatality and injured rates (Wyss & Trendafiloski, 2011). Unfortunately, the casualty reports after earthquakes almost never specify the minimum degree of injury required to be listed. The occupancy rates in QLARM were assumed to corresponding to the earthquake occurrence hour. The maximum occupancy rate of 99% represents nighttime and is used in the assumed worst case scenarios. In the daytime hours, the minimum occupancy is taken to be 26%. In terms of the assumption of the occupancy rates, human losses are calculated using the casualty event-tree model. The probability of casualty state c_i for seismic intensity I is therefore estimated as a product of the damage probabilities in I and the casualty

probabilities in damage degree D_k (for more detail, see Trendafiloski et al., 2011):

$$P(c_i I_j) = \sum_{k=1}^{3} P\,(D_k I_j) P(D_k c_i) + P(D_{NC} I_j) P(D_{NC} c_i) + P(D_c I_j) P(D_C c_i) \tag{3}$$

In which

$$P(D_c I_j) = k_c(I_j)[P(D_4 I_j) + P(D_5 I_j)]$$

$$P(D_{NC} I_j) = [1 - k_c(I_j)][P(D_4 I_j) + P(D_5 I_j)] \tag{4}$$

where $P(D_k I_j)$ is the probability of occurrence of damage grades k = 1 to 3 for seismic intensity I_j; $P(D_{NC} I_j)$ is the probability of having no collapse among the buildings with damage grades 4 and 5; $P(D_C I_j)$ is the probability of having collapse among the buildings with damage grades 4 and 5; $k_C(I_j)$ is the collapse model; $P(D_k c_i)$ is the probability of having casualty state c_i due to damage grade D_k. QLARM calculates these casualties for each settlement.

We have confidence in the sum total of casualties over all affected settlements, which is our final assessment of the extent of an earthquake caused disaster. This is so because uncertainties that affect estimates in single settlements, such as ground condition, average out in the sum over thousands of settlements, each with thousands of buildings on various soils. However, estimates of casualties for specific single settlements are not reliable because of the many unknown that affect such a specific calculation. QLARM only considers calculations of the human losses due to expected damage caused by ground shaking. It does not consider other sources of fatalities, such as tsunamis, landslides or earthquake-related fires. Details on the population and building data of the world are contained in a number of reports on the website of the ICSE Foundation (http://www.icesfoundation.org/Pages/CustomPage.aspx?ID=4, last accessed 2 April 2018). Much information about common building types and occupancy rate has been collected by the World Housing Encyclopedia (WHE) (http://www.world-housing.net/, last accessed 2 April 2018), especially in the WHE/PAGER project.

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